

Digital Material Passport

Manufacturer		Customer	Business Transaction	
Nordstahl Schmiedewerke GmbH Werkstraße 12 44135 Dortmund DE info@nordstahl-schmiede.example.com		ACME Steel Trading GmbH Handelsweg 5c 76199 Karlsruhe DE customer@example.com	Order	PO-2025-8842 · 19000 kg
			Delivery	DN-2025-8842 · 3800 kg

Product

Product Name	Forged round bars, rough turned	Name (EN 10027-1)	18CrNiMo7-6 +FP
Batch ID	BATCH-2025-8842	Number (EN 10027-2)	1.6587
Surface Condition	Rough turned	Shape	RoundBar - D 500 mm - Length 1500 mm
Delivery Condition	+FP - (Ferrite/pearlite annealed)	Marking	Head of bar stamped with grade, size, heat No., weight, forging No. Marking has to allow clear identification
Weight	9.5 t	Packaging	Bulk, max. weight of bar 13,5 mt
Diameter Tolerance	+3/-0 mm	Coloring	Yellow/black
Specification	8842-1/ GENERIC - FRT - Rev 1.0 - 2025-03-18	Product Norm	ISO 683-3
		Product Norm	ISO 6336-5

Heat Treatment

Process: **Annealing** - Lot: HT-LOT-2025-8842 - Furnace: F-12 - Date: 2025-03-10

Stage	Temperature	Duration	Cooling	Atmosphere
Soaking	650 C	6 h		Air

Chemical Analysis

Heat Number **HT-2025-4521** - Melting Process **EAF+LF+VD** - Casting Method **ContinuousCasting** - Sample Location **Ladle**

Symbol	C	Si	Mn	P	S	Cr	Ni	Mo	Cu	Ti	Al	V	Sn	N	Ca	O	H	F1	F2
Unit	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	ppm	%	-
Min	0.15	0.15	0.5			1.5	1.4	0.25			0.02			0.01					2
Max	0.21	0.4	0.9	0.02	0.008	1.8	1.7	0.35	0.25	0.005	0.05		0.03	0.017	0.0025	25	2	0.5	3.5
Actual	0.18	0.27	0.65	0.01	0.004	1.65	1.55	0.3	0.12	0.003	0.03	0.005	0.012	0.013	0.0015	12	1.2	0.24	2.31

F1 = Cu+10*Sn: 0.24% F2 = Al/N: 2.31-

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method
Tensile Strength ¹ Specimen: 1/4T, L	Rm	1265 N/mm²	1200 N/mm²		
Yield Strength ¹ Specimen: 1/4T, L	Rp0,2	905 N/mm²	850 N/mm²		
Elongation ¹ Specimen: 1/4T, L	A5	13.5 %			
Reduction of area ¹ Specimen: 1/4T, L	Z	52 %			
Impact test ISO-V +20°C on blank hardened sample (#30 mm), 870°C +/-10°C 2h / water / 180°C -0/+10°C 4h / air cooling Specimen: 1/4T, L		58 / 62 / 60 J - Average 60 - Min 58	40 J		
Impact test ISO-V -30°C on blank hardened sample (#30 mm), 870°C +/-10°C 2h / water / 180°C -0/+10°C 4h / air cooling Specimen: 1/4T, L		45 / 48 / 47 J - Average 46.7 - Min 45			
Surface hardness		185 HB	159 HB	207 HB	
Core hardness		178 HB	159 HB	207 HB	

¹ on blank hardened sample (#30 mm), 870°C +/-10°C 2h / water / 180°C -0/+10°C 4h / air cooling

Physical Properties

Property	Actual	Target/Min	Maximum	Method
Grain size	7	6		ISO 643
Purity				ISO 4967 method B
Plate II, inspected area approx. 200 mm²				

Inclusion type	A fine	A thick	B fine	B thick	C fine	C thick	D fine	D thick	DS
Value	2	1.5	1.5	1	1	0.5	1.5	1	1
Max	3	3	2.5	1.5	2.5	1.5	2	1.5	2

Purity K4	14					22		DIN 50602
Straightness	1.5 mm/m					2.5 mm/m		
Reduction ratio	6			4				
Residual magnetism	8 A/cm					15 A/cm		
Structure	Ferrite/pearlite							

✓ Fully formed structure ✓ Protected from reoxidation

Supplementary Tests

Test	Result / limits	Method
Ultrasonic test 100%	Yes - Pass	EN 10228-3 Type 1a class 3
Surface test 100% visually checked, clean surface free from defects	Yes - 100% visually checked, clean surface free from defects	
Radioactivity	Yes - max 100 Bq/kg	
Square cut obliquity max. 2% of diameter, no centerholes	Yes - Max. 2% of diameter, no centerholes	

✓ Free from mercury ✓ Free from cadmium ✓ 100% checked ✓ Sawn, free from burr

Hardenability – Jominy (ISO 642), values in HRC - +HH band; tested per ISO 642 or calculated values

Distance - from - quenched - end (mm)	1.5	3	5	7	9	11	13	15	20	25	30	35	40	45	50
Value [HRC]	46	46	45	44	44	44	42	42	40	39	38	37	37	30	30
Min	43	43	42	41	40	40	39	38	36	35	34	33	33		
Max	48	48	48	48	47	47	46	46	44	43	42	41	41		

Validation

We confirm that the goods described above comply with the requirements of the order and this inspection document was prepared in accordance with EN 10204:2004, type 3.1.

- ✓ 100% of the material is from Chinese sources
- ✓ Material is of non-Russian origin (EU Regulation No. 833/2014)



Validated by **Dr. Lena Krause**, Head of Quality Assurance, Quality Assurance - 2025-03-18